

Forces and motion

Scalar: A quantity with magnitude only

Examples:

- * Distance
- * Speed
- * Mass
- * Volume
- * Density
- * Temperature
- * Energy / Work

Vector: A quantity with magnitude and direction

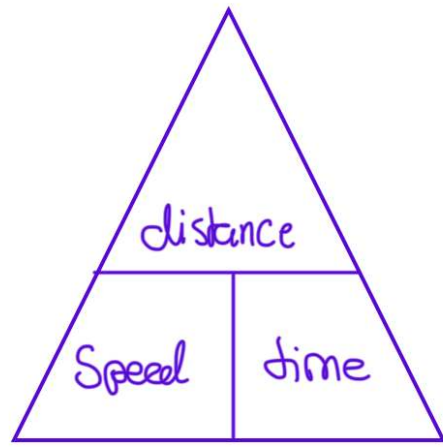
Examples:

- * Displacement
- * Velocity
- * Weight
- * Force
- * Acceleration
- * Momentum
- * Moment
- * Gravitational field strength
- * Electric current

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} \rightarrow \text{m/s}$$

$$\text{Distance} = \text{Speed} \times \text{time} \rightarrow \text{m}$$

$$\text{time} = \frac{\text{Distance}}{\text{Speed}} \rightarrow \text{s}$$



Velocity can be positive or negative to show direction

+ $\rightarrow$ right (forward)

- $\leftarrow$ left (backward)

Convert from

a) kg/hr to m/s

b) m/s to kg/hr

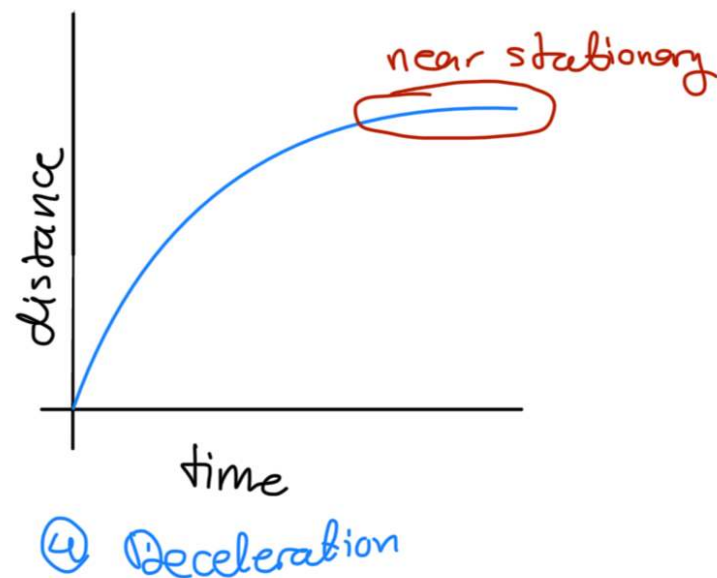
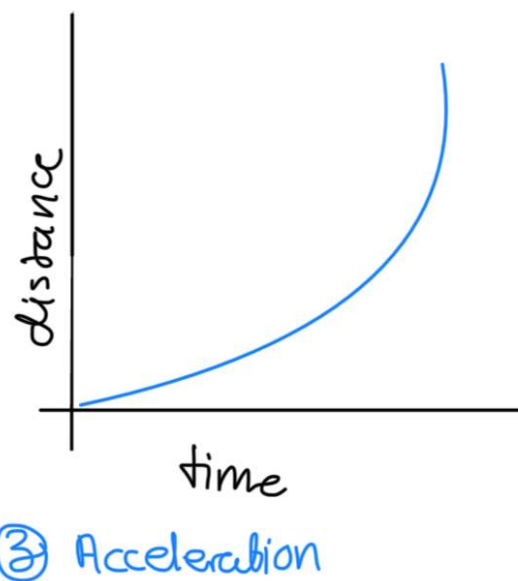
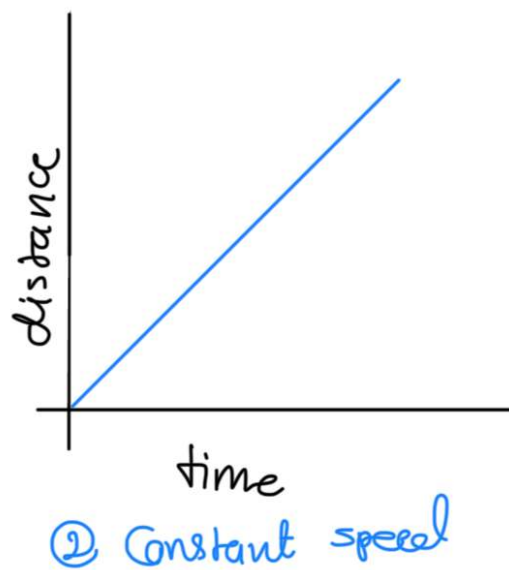
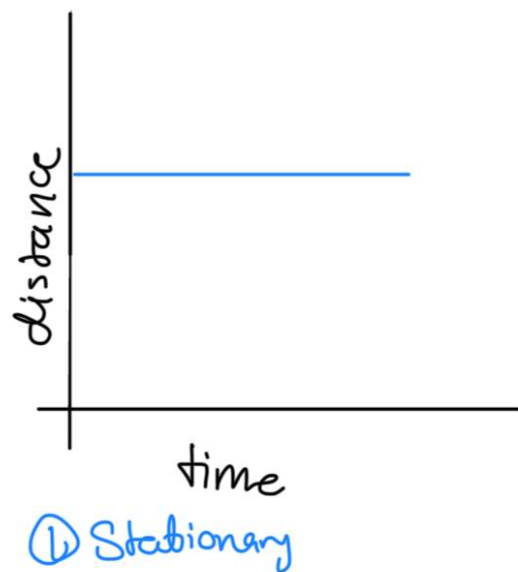
$$\frac{\text{kg}}{\text{hr}} \times \frac{1000}{3600}$$

$$\frac{\text{m}}{\text{s}} \div \frac{1000}{3600} \quad \frac{\text{m}}{\text{s}} \times \frac{3600}{1000}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{total time}}$$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{time}} = \frac{\text{Final} - \text{initial}}{\text{time}} = \frac{v - u}{t} \rightarrow \text{m/s}^2$$

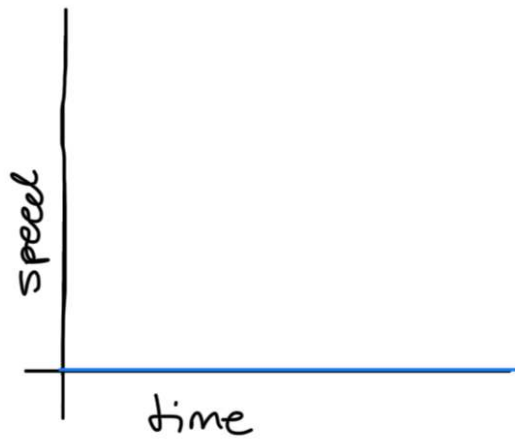




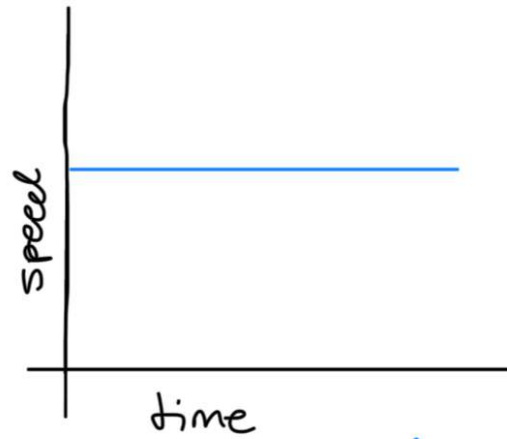
Gradient: $\frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Distance}}{\text{time}} = \text{Speed}.$

Area under the graph represents nothing.

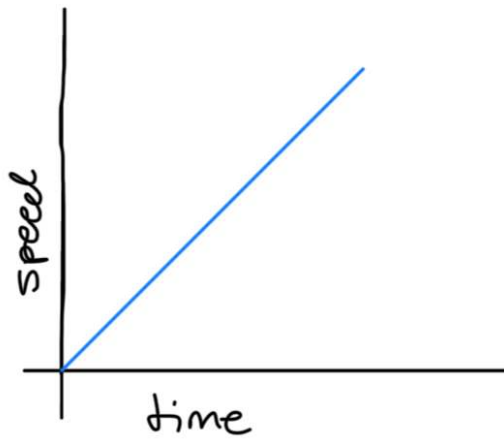
Speed - time graph



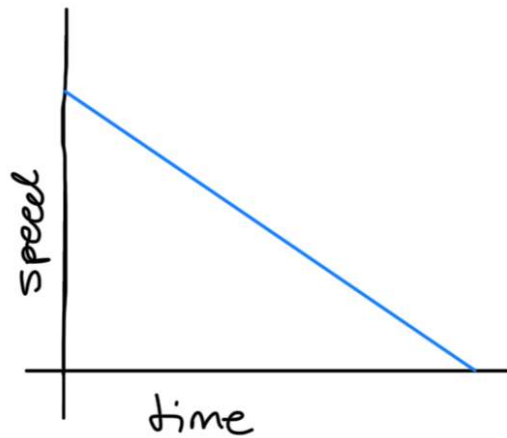
① Stationary



② Constant speed



③ Acceleration

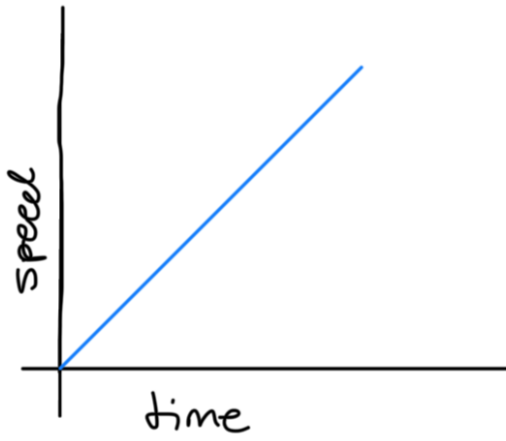


④ Deceleration

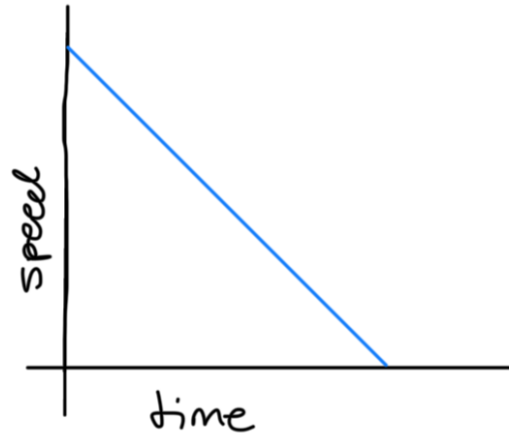
Gradient: $\frac{\text{Speed}}{\text{time}}$: acceleration

Area under the graph: Distance

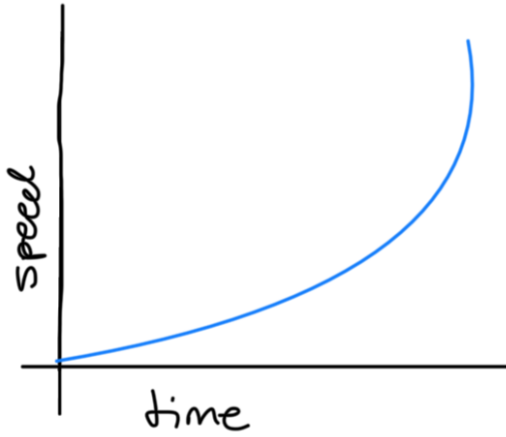
Uniform acceleration:



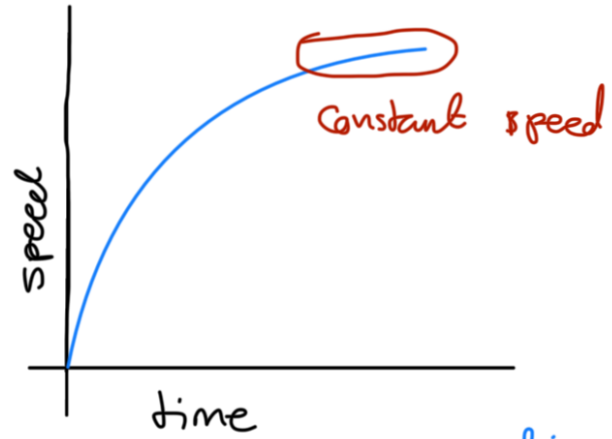
① Acceleration



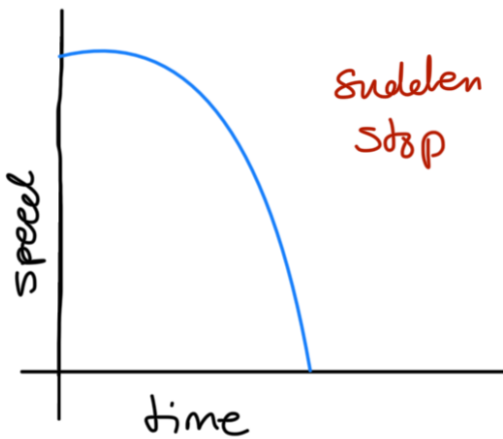
② Deceleration



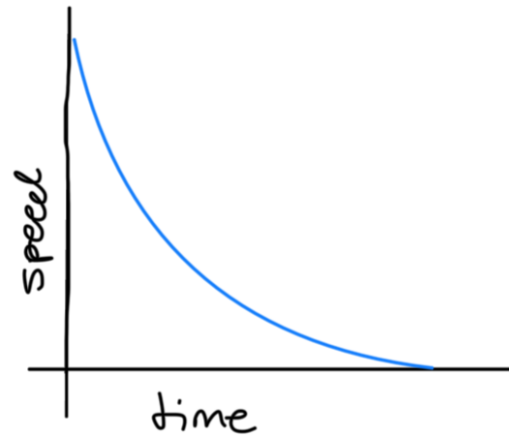
③ Increasing acceleration



④ Decreasing acceleration

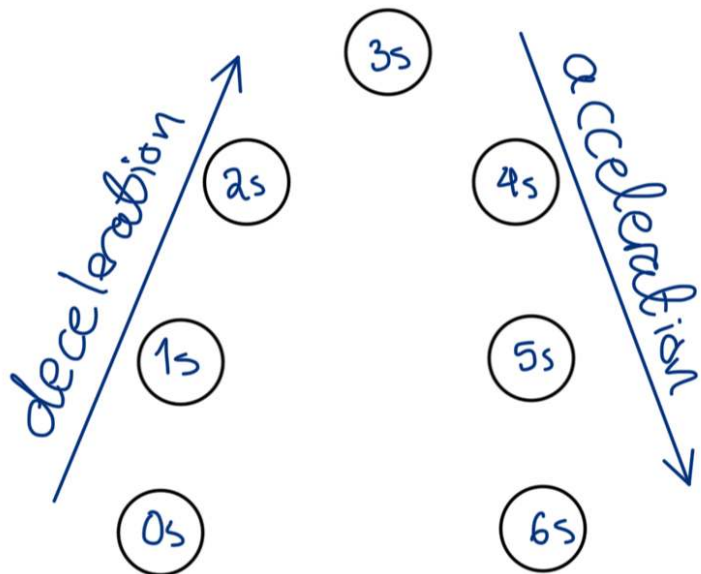
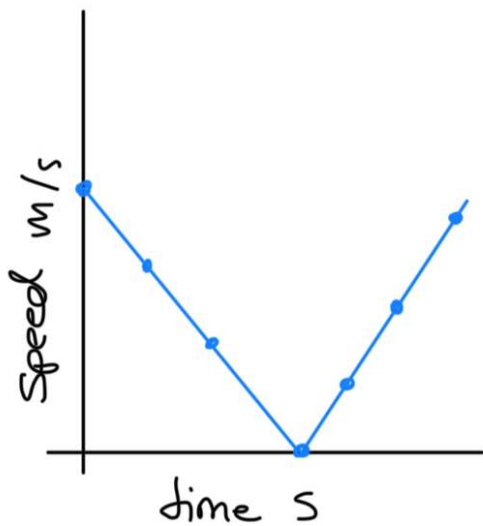
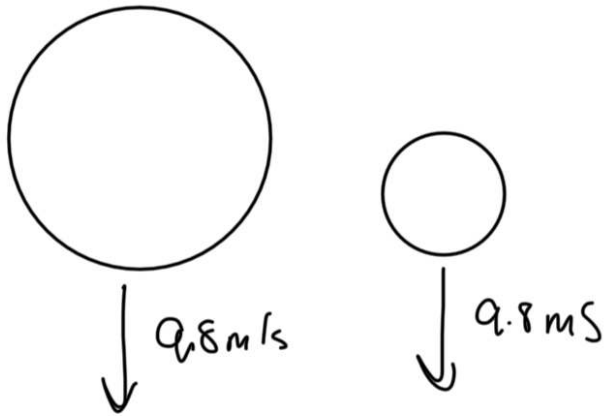


⑤ Increasing deceleration



⑥ Decreasing deceleration

If two objects, with different weights, are dropped from a high point, without air resistance acting on them, only gravity, both of them will reach the ground at the same time. This is because both objects are pulled down by the same force = gravitational pull = gravitational acceleration
gravitational acceleration = 9.8 m/s^2



Forces

Contact force

- * Push / Pull
- * Friction
- * Air resistance
- * Thrust
- * Upthrust
- * Buoyancy
- * Tension
- * Normal
- * Drag

Act-on-a distance force

- * Gravity
- * Magnetic
- * Electric

Newton's First law of motion:

For objects in equilibrium, an object at rest remains at rest unless affected by an external force. An object moving at constant speed will continue to move at constant speed unless affected by an external object.

$$F_{\text{net}} = 0$$

Newton's Second law of motion:

Force = mass $\times$ acceleration

$$F = ma \rightarrow \text{m/s}^2$$

$\swarrow$ N $\downarrow$ kg

$$F_{\text{net}} \neq 0$$

① Momentum : mass $\times$ velocity

$$P = m \times v$$

$\text{kgm/s} \leftarrow \text{kg} \leftarrow \text{m/s}$

- : left direction

+ : Positive direction -

$$\rightarrow \Delta P = P_2 - P_1$$

$$\rightarrow \Delta P = F \times \Delta t \rightarrow \text{Impulse : kgm/s / N.s}$$

$$\rightarrow F = \frac{\Delta P}{\Delta t} = \frac{mv - ma}{t}$$

② Conservation of momentum :

Total momentum before = Total momentum after.

* Without external force.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

(before) (after)

→ Equal Force for both objects

→ Equal time of contact for both objects

→ Equal impulse.

Case 1: Two balls before , two balls after

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

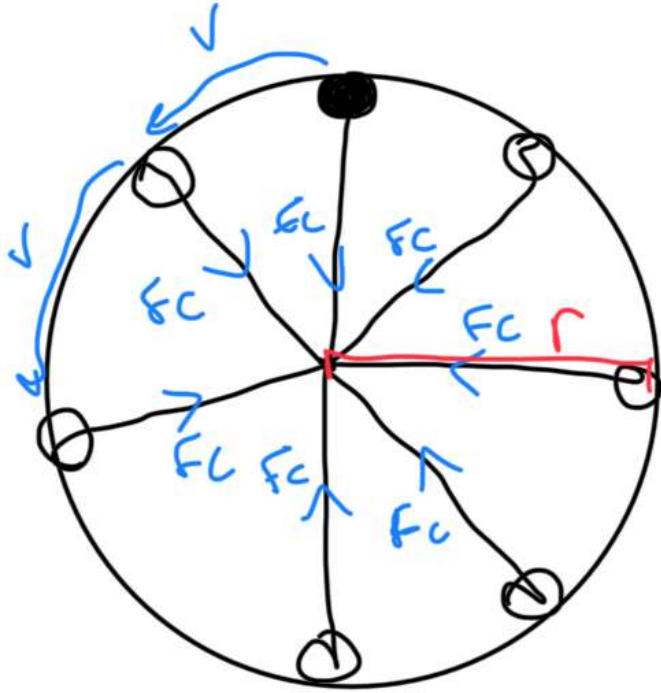
Case 2: 2 balls before , only 1 after

$$m_1 u_1 + m_2 u_2 = v(m_1 + m_2)$$

Case 3: 1 ball before , two after

$$m u = m_1 v_1 + m_2 v_2$$

In circular motion, the object is moving at constant speed and accelerating at the same time.



F_c : Centripetal force
"Center seeking force"
= Tension

$F_c \propto \text{velocity } (v)$

$F_c \propto \frac{1}{\text{radius } (r)}$

$F_c \propto \text{mass}$

$F_c \propto \frac{\text{velocity} \times \text{mass}}{\text{radius}}$

$$F_c = \frac{m \times v^2}{r}$$

① Extension : Total new length - Total original length.

② Force is directly proportional to extension
 $F \propto x$

③ Hooke's Law : $F = kx$
 $\begin{matrix} \nearrow N/m \\ N \downarrow \end{matrix} \quad \begin{matrix} \nearrow \\ L \rightarrow m \end{matrix}$

④ $\uparrow k \uparrow = \frac{F \uparrow}{x \downarrow}$
 $\rightarrow k$ is directly proportional to F
 $\rightarrow k$ is inversely proportional to x

⑤ For greater constant (k):
- Larger force
- Less extension

⑥ Limit of proportionality: when the spring stops obeying Hooke's law

